

ECS 189G Spring 2026: Prompting with In-Context Learning and Evaluation

TO PREPARE AND SUBMIT HOMEWORK

Follow these steps exactly, so your assignment can be properly graded:

1. Carefully read the README.MD in the code.
2. Complete your evaluation scripts (e.g., `eval_sst2.py`) demonstrating dynamic prompt building and your custom text parsing logic.
3. Compile your manual observations, the k-shot accuracy plot, and your prompt sensitivity analysis into a single PDF file named `Report.pdf`.
4. Package your Python scripts, any modified starter code, and your `Report.pdf` into a single .zip archive.
5. Upload your .zip file to the course submission portal by the due date.

INSTRUCTIONS

In this assignment, you will explore the power and limitations of **In-Context Learning (ICL)**. You will design prompts to guide a base Large Language Model (**Qwen3-0.6B-Base**) to perform sentiment analysis on the **SST-2 (Stanford Sentiment Treebank)** dataset.

The goal is to understand how prompting and ICL affect performance and how to evaluate results rigorously under variance and prompt sensitivity. Because base models do not inherently stop after answering, you will face the "Parsing Challenge"—implementing programmatic logic to extract exact labels from a continuous output stream.

- **Model:** Qwen3-0.6B-Base (CPU-friendly, but will use GPU if available)
- **Dataset:** SST-2 provided in JSON format (`train.json`, `val.json`).
- **Task:** Binary sentiment classification (0 = Negative, 1 = Positive).

ENVIRONMENT SETUP

It is highly recommended to use a virtual environment (e.g., Conda or Python venv) to avoid dependency conflicts. We have provided a `requirements.txt` file for you.

PIPELINE IMPLEMENTATION & PARSING [45 POINTS]

You must build a systematic evaluation pipeline to handle the unique behaviors of base models (e.g., create a new script `eval_sst2.py`):

- **[25 POINTS] Dynamic k-shot Prompting:** Write a function to dynamically sample **K** exemplars from the training set and format them into a prompt template for validation queries.
- **[20 POINTS] The Parsing Challenge:** Since the base model will continue generating tokens beyond the answer, implement a robust text parser (e.g., using string manipulation or truncation) to extract only the target label ("Positive" or "Negative").

ICL SCALING ANALYSIS (K-SHOT) [45 POINTS]

Investigate how the number of in-context examples (K) impacts classification accuracy.

- **[25 POINTS] Execution:** Run your evaluation loop across K in $\{0, 1, 3, 5, 8, 16\}$ shots.
- **[20 POINTS] Analysis:** Plot an Accuracy vs. k -shot curve. In your report, discuss how the model utilizes increasing exemplars to "learn" the task pattern.

ROBUSTNESS & SENSITIVITY ANALYSIS [10 POINTS]

Base models are notoriously brittle to minor formatting changes. Fix your exemplar count (e.g., $k=3$ or 5) and test the following:

- **Format Shift:** Compare different structural layouts (e.g., Review: [Text] \n Sentiment: vs. Input: [Text] | Label:).
- **Label Word Shift:** Evaluate the impact of swapping target mapping words (e.g., Positive/Negative vs. Good/Bad or 1/0).
- **Discussion:** Discuss in your report why the model's performance fluctuates and how sensitive it is to these variations.